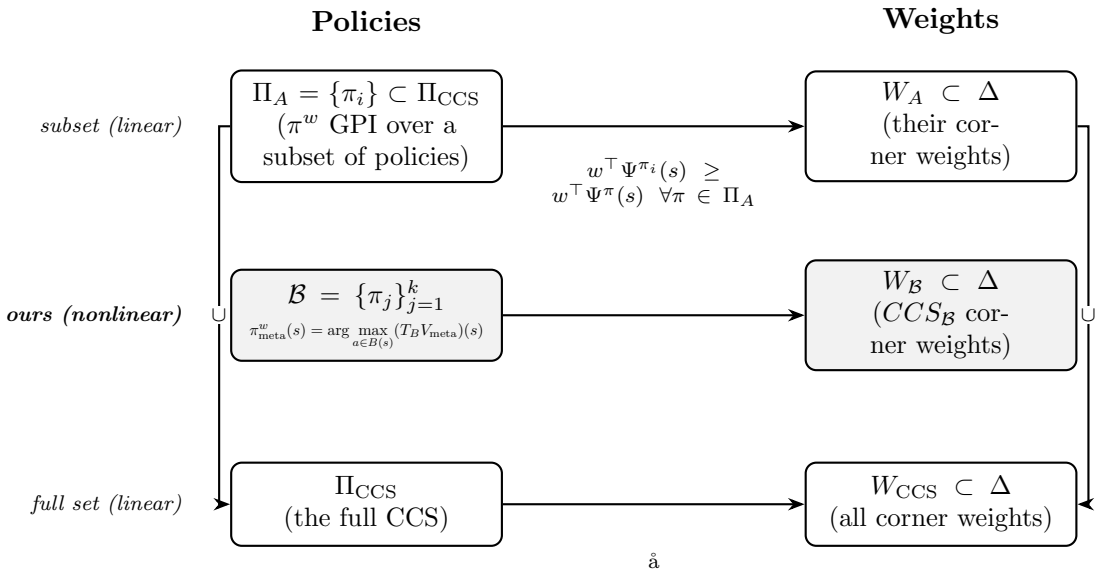
Certified Policy Bases for Successor-Feature Composition

Overview:



In rows 1 and 3 the relation is linear in set union: $W_{A \cup B} = W_A \cup W_B$, so adding a policy only ever adds its own corner weights, never changes anyone else's.

1 Goals for a publication

- Compute the CCS faster by using the combination of base and meta policies. Note, however, that it is not enough to get all the policies from the CCS, but it is more important to get the successor features to perform GPI correctly.
- Provide an improved version of GPI, when the CCS is very big that we don't have time to compute all the policies, then perform GPI on a meta-policy particularly computed for that task.
- Look how to implement it in Infante's paper.

2 Setup

A MO MDP is $\mathcal{M} = (S, A, P, \phi, \gamma)$ with vector reward $\phi : S \times A \to \mathbb{R}^d$. For a weight $w \in \Delta^{d-1} = \{w \geq 0,\ ||\mathbf{w}||_1 = 1\}$, the scalarized reward is $\phi(s,a)^\top w$. The successor feature (SF) of policy π is

$$\Psi^\pi(s,a) = \mathbb{E}_\pi\Big[\sum_{t\geq 0}\gamma^t\phi(s_t,a_t)\ \Big|\ s_0 = s, a_0 = a\Big], \qquad \Psi^\pi(s,a) = \phi(s,a) + \gamma\,\mathbb{E}_{s'}\big[\Psi^\pi(s',\pi(s'))\big].$$

$V^*_w(s) = \max_\pi \Psi^\pi(s,\pi(s))^\top w$ is the scalarized optimal value. The *convex coverage set* (CCS) is the set of SF vectors of policies optimal for some w; recovering it is the standard multi-objective planning goal (Optimistic Linear Support, OLS).

Definition 1 (Basis)**.** *A* basis *is a finite set $\mathcal{B} = \{(\pi_j, \Psi^{\pi_j})\}_{j=1}^k$ of exactly solved policies, each π_j optimal for its own training weight w_j: $\pi_j(s) = \arg\max_a \Psi^{\pi_j}(s,a)^\top w_j$. Each π_j is fixed once solved.*

Definition 2 (Restricted MORL)**.** *For a basis $\mathcal{B}$ and a state s, the restricted action set is exactly the base policies' own actions,*

$$B(s) = \{\pi_1(s), \ldots, \pi_k(s)\} \subseteq A(s), \qquad \textit{i.e.}\quad a \in B(s) \iff a = \pi_j(s) \text{ for some } j.$$

For a query weight w, the restricted Bellman operator is

$$(T_B V)(s) = \max_{a\in B(s)}\big[\phi(s,a)^\top w + \gamma\,\mathbb{E}_{s'}V(s')\big] = \max_{j\in\{1,\ldots,k\}}\big[\phi(s,\pi_j(s))^\top w + \gamma\,\mathbb{E}_{s'}V(s')\big],$$

a γ-contraction with unique fixed point

$$V_{\text{meta}} = T_B V_{\text{meta}},$$

We define a metapolicy over $\mathcal{B}$ as the greedy policy of this fixed point:

$$\pi^w_{\text{meta}}(s) \ =\ \pi_{j^*(s)}(s), \qquad j^*(s) = \arg\max_{j\in\{1,\ldots,k\}}\big[\phi(s,\pi_j(s))^\top w + \gamma\,\mathbb{E}_{s'}V_{\text{meta}}(s')\big],$$

Definition 3 (Bellman certificate)**.** *For candidate value V and weight w, the gap against the unrestricted operator T is*

$$\text{gap}_w(s;V) = (TV)(s) - V(s) = \max_{a\in A(s)}\big[\phi(s,a)^\top w + \gamma\,\mathbb{E}_{s'}V(s')\big] - V(s).$$

Note that $\text{gap}_w(s;V_{\text{meta}})$ is always ≥ 0 for any s because $B(s) \subseteq A(s)$. If $\text{gap}_w(s;V_{\text{meta}}) = 0$ for all s, then $V_{\text{meta}} = V^*_w$, and the metapolicy is optimal for that weight.

Algorithm 1 CERTIFY-OR-GROW$(w, \mathcal{B})$

$\pi_{\text{meta}}, \Psi^{\pi_{\text{meta}}} \leftarrow$ restricted policy iteration on $\mathcal{B}$ at w ▷ full table, not just V_{meta}: one extra vector solve for the fixed greedy policy
$V_{\text{meta}}(\cdot) \leftarrow \Psi^{\pi_{\text{meta}}}(\cdot, \pi_{\text{meta}}(\cdot))^\top w$
if $\max_s \text{gap}_w(s; V_{\text{meta}}) \leq \text{tol}$ **then**
    return $\pi_{\text{meta}},\ \Psi^{\pi_{\text{meta}}},\ \mathcal{B}$
else
    $(\pi_w, \Psi^{\pi_w}) \leftarrow$ unrestricted policy iteration at w
    $\mathcal{B} \leftarrow \mathcal{B} \cup \{(\pi_w, \Psi^{\pi_w})\}$ ▷ π_w is trivially $\mathcal{B}$'s new metapolicy: gap 0 by construction
    return $\pi_w,\ \Psi^{\pi_w},\ \mathcal{B}$
end if

2.1 How can we improve the GPI from base policies?

The restricted metapolicy π_{meta} might not beat GPI, since GPI can also take actions no base policy would ever take. We now motivate a policy that does at least as well.

Definition 4 (GPI estimate)**.** *For a policy set* $\Pi = \{\pi_1, \dots, \pi_k\}$ *and weight* w, $\widehat{Q}_\Pi(s,a) := \max_j \Psi^{\pi_j}(s,a)^\top w$, *and* $\pi_\Pi^{\text{GPI}}(s) := \arg\max_a \widehat{Q}_\Pi(s,a)$.

By the SF Bellman equation, for every j and *every* action a (not only $\pi_j(s)$),

$$\Psi^{\pi_j}(s,a)^\top w = \phi(s,a)^\top w + \gamma\, \mathbb{E}_{s' \sim P(\cdot|s,a)}\big[V^{\pi_j}(s')\big], \qquad V^{\pi_j}(s) := \Psi^{\pi_j}(s,\pi_j(s))^\top w.$$

The instant-reward term does not depend on j, so $\max_j$ passes through it:

$$\widehat{Q}_{\mathcal{B}}(s,a) \;=\; \phi(s,a)^\top w + \gamma \max_j \mathbb{E}_{s'}\big[V^{\pi_j}(s')\big]. \tag{$*$}$$

Lemma 1 (V_{meta} dominates every base policy's own value)**.** $V_{\text{meta}}(s) \geq V^{\pi_j}(s)$ *for every* j, s.

Substituting V_{meta} for the future-reward term in $(*)$ can only grow it, and by the Bellman equation for the fixed policy π_{meta} equals $\Psi^{\pi_{\text{meta}}}(s,a)^\top w$:

$$\widehat{Q}_{\mathcal{B}}(s,a) \;\leq\; \phi(s,a)^\top w + \gamma \mathbb{E}_{s'}[V_{\text{meta}}(s')] \;=\; \Psi^{\pi_{\text{meta}}}(s,a)^\top w \qquad \text{for every } (s,a). \tag{$**$}$$

Theorem 1 (π^{hyb} matches the pool estimate, from V_{meta} alone)**.** *Let* $\pi^{\text{hyb}}(s) := \arg\max_a \big[\phi(s,a)^\top w + \gamma\, \mathbb{E}_{s'} V_{\text{meta}}(s')\big]$. *Then*

$$Q^{\pi^{\text{hyb}}}(s,a) \;\geq\; \widehat{Q}_{\mathcal{B}}(s,a) \qquad \textit{for every } (s,a).$$

Hence, we have two options, we do GPI and then we perform some iteration to improve our estimate is particular places. Or we can do GPI, and compute the metapolicy as well normally.

To obtain the successor vectores we use the following algorithm:

Algorithm 2 MetaPolicyQL($\{\pi_1, \dots, \pi_k\}, w$) — metapolicy and its SF table, by sampling

$\widehat{\Psi}(s,a) \leftarrow \mathbf{0}$ for every (s,a)
repeat
  observe $(s, a, \phi(s,a), s')$ under any behavior policy with adequate state-action coverage
  $\hat{\jmath} \leftarrow \arg\max_{j \in \{1,\dots,k\}} \widehat{\Psi}\big(s', \pi_j(s')\big)^\top w$ ▷ meta-greedy action at s', restricted to the k base policies' own actions
  target $\leftarrow \phi(s,a) + \gamma\, \widehat{\Psi}\big(s', \pi_{\hat{\jmath}}(s')\big)$
  $\widehat{\Psi}(s,a) \leftarrow \widehat{\Psi}(s,a) + \alpha\, \big[\text{target} - \widehat{\Psi}(s,a)\big]$
until converged
$\pi_{\text{meta}}(s) \leftarrow \pi_{j^*(s)}(s)$, $j^*(s) := \arg\max_j \widehat{\Psi}(s, \pi_j(s))^\top w$, for every s
return π_{meta}, $\widehat{\Psi}$

Definition 5 (Vector lift of T_B)**.** *For* $\Psi : S \times \{1, \dots, k\} \to \mathbb{R}^d$,

$$(\mathcal{T}_B \Psi)(s,j) := \phi(s, \pi_j(s)) + \gamma\, \mathbb{E}_{s'}\big[\Psi(s', \hat{\jmath}(s'))\big], \qquad \hat{\jmath}(s') := \arg\max_{j'} \Psi(s', j')^\top w.$$

Remark 1 (Not unconditionally a contraction)**.** *For* $\Psi_1 \neq \Psi_2$, $(\mathcal{T}_B\Psi_1)(s,j) - (\mathcal{T}_B\Psi_2)(s,j)$ *involves* $\Psi_1(s', \hat{\jmath}_1(s')) - \Psi_2(s', \hat{\jmath}_2(s'))$ *at possibly* different *maximizers* $\hat{\jmath}_1 \neq \hat{\jmath}_2$*; nothing bounds this by* $\|\Psi_1 - \Psi_2\|_\infty$ *in general — the same mixing of a scalarized* arg max *with a full vector quantity that broke Theorem 3's predecessor.* $\mathcal{T}_B$ *is* not *a contraction on* $(\mathbb{R}^d)^{S \times \{1,\dots,k\}}$*; the correct argument is below.*

Lemma 2 (Scalarization reproduces T_B exactly)**.** *Let $V_n(s) := \max_j \Psi_n(s,j)^\top w$ for the iterates $\Psi_{n+1} = \mathcal{T}_B \Psi_n$. Then $V_{n+1} = T_B V_n$ for every n.*

Proof. $\hat{\jmath}(s')$ maximizes $\Psi_n(s',\cdot)^\top w$, so $\Psi_n(s',\hat{\jmath}(s'))^\top w = V_n(s')$ exactly; dotting Definition 5 with w gives $(\mathcal{T}_B\Psi_n)(s,j)^\top w = \phi(s,\pi_j(s))^\top w + \gamma \mathbb{E}_{s'}[V_n(s')]$, so $V_{n+1}(s) = \max_j (\mathcal{T}_B\Psi_n)(s,j)^\top w = (T_B V_n)(s)$. □

Lemma 3 (Finite-time policy stabilization)**.** *If, for every s, the two largest values of $j \mapsto \phi(s,\pi_j(s))^\top w + \gamma \mathbb{E}_{s'} V_{\text{meta}}(s')$ are distinct, there is N such that $\hat{\jmath}_n(s) := \arg\max_j \Psi_n(s,j)^\top w$ equals π_{meta}'s own greedy index at every s, for all $n \geq N$.*

Proof. By Lemma 2 and T_B's contraction (Definition 2), $V_n \to V_{\text{meta}}$ geometrically, so $\Psi_n(s,j)^\top w \to Q_B(s,j) := \phi(s,\pi_j(s))^\top w + \gamma \mathbb{E}_{s'} V_{\text{meta}}(s')$ uniformly in j. Once this error is below half the action gap at s, the ordering of $\Psi_n(s,\cdot)^\top w$ matches $Q_B(s,\cdot)$'s, so $\hat{\jmath}_n(s)$ matches Q_B's maximizer — exactly π_{meta}'s own greedy index (Definition 2). □

Theorem 2 (The merged exact recursion converges to $(\pi_{\text{meta}}, \Psi^{\pi_{\text{meta}}})$)**.** *Under Lemma 3's non-degeneracy condition, $\Psi_n(s, j^*(s)) \to \Psi^{\pi_{\text{meta}}}(s, \pi_{\text{meta}}(s))$ for every s.*

Proof. For $n \geq N$ (Lemma 3), $\hat{\jmath}_n \equiv j^*$ is constant, so $\Psi_{n+1}(s,j) = \phi(s,\pi_j(s)) + \gamma \mathbb{E}_{s'}[\Psi_n(s', j^*(s'))]$ has no further branching; restricted to $j = j^*(s)$ this is exactly the Bellman equation for the fixed policy π_{meta}, a γ-contraction in the vector sup-norm with unique fixed point $\Psi^{\pi_{\text{meta}}}(s, \pi_{\text{meta}}(s))$. □

This is the reliability guarantee: π^{hyb}, built from V_{meta} alone, matches GPI's own pool *estimate* everywhere — the same guarantee GPI gives about itself, at the cost of Remark 3 instead of a full per-action pool. It is *not* a claim about GPI's own *realized* value $Q^{\pi_{\mathcal{B}}^{\text{GPI}}}$, which is generally larger than its own estimate $\widehat{Q}_{\mathcal{B}}$ — Theorem 3 next shows neither π^{hyb} nor $\pi_{\mathcal{B}}^{\text{GPI}}$ dominates the other's realized value in general.

Theorem 3 (The tie fails for $k \geq 2$, in both directions)**.** *For $k \geq 2$, $Q^{\pi^{\text{hyb}}} \neq Q^{\pi_{\mathcal{B}}^{\text{GPI}}}$ in general. There exist finite deterministic MDPs, bases $\mathcal{B} = \{\pi_1, \pi_2\}$, and query weights w exhibiting this in both directions — π^{hyb} strictly ahead of $\pi_{\mathcal{B}}^{\text{GPI}}$ in one example, strictly behind in another — so neither dominates the other in general.*

Example (π^{hyb} strictly better). $\gamma = 0.9$; states $\{0, 1, 2\}$, actions $\{0, 1, 2\}$, deterministic $\delta(s, a)$ and $\phi(s, a) \in \mathbb{R}^2$:

	$a = 0$	$a = 1$	$a = 2$
$s = 0$	δ=0, ϕ=(.198, .761)	δ=0, ϕ=(.727, .507)	δ=1, ϕ=(.276, .218)
$s = 1$	δ=2, ϕ=(.703, .560)	δ=0, ϕ=(.532, .130)	δ=0, ϕ=(.266, .699)
$s = 2$	δ=1, ϕ=(.983, .066)	δ=1, ϕ=(.728, .631)	δ=1, ϕ=(.751, .081)

$\mathcal{B} = \{\pi_1, \pi_2\}$, $\pi_1 = (1, 0, 1)$, $\pi_2 = (2, 0, 0)$, query $w = (0.152, 0.848)$. Then $B(0) = \{1, 2\}$, $B(1) = \{0\}$, $B(2) = \{0, 1\}$; restricted PI gives $\pi_{\text{meta}} = (2, 0, 1)$, $V_{\text{meta}} = (5.7376, 6.1232, 6.1569)$, and here $\pi_{\mathcal{B}}^{\text{GPI}} = \pi_{\text{meta}}$ exactly, so $Q^{\pi_{\mathcal{B}}^{\text{GPI}}}(0, \cdot)$'s value is 5.7376. Action $0 \notin B(0)$ is a self-loop ($\delta(0, 0) = 0$) invisible to the metapolicy, yet $\phi(0, 0)^\top w + \gamma V_{\text{meta}}(0) = 0.6749 + 0.9(5.7376) = 5.8386 > V_{\text{meta}}(0)$, so $\pi^{\text{hyb}}(0) = 0$, realizing $\phi(0, 0)^\top w/(1 - \gamma) = 6.7489 > 5.7376$. GPI misses this because its pool estimate for action 0 bootstraps off *each base policy's own* continuation ($\max(\Psi^{\pi_1}(0, 0)^\top w, \Psi^{\pi_2}(0, 0)^\top w) = \max(5.542, 4.151) = 5.542 < 5.7376$), never off the already-composed V_{meta} that π^{hyb} has access to. □

Remark 2 (Neither direction is rare, nor structural)**.** *A 3×4×2 example (`SFOLS/08_hyb_vs_gpi_conjecture.ipyn`) gives the reverse ranking, $V^{\pi_{\mathcal{B}}^{\text{GPI}}}(0) = 7.013 > V^{\pi^{\text{hyb}}}(0) = 6.601$, from a single mistaken choice by π^{hyb} one step downstream of where the two policies first disagree — so neither direction of Theorem 3 is an artifact of one mechanism. Adding an absorbing zero-reward sink on every pickup (mirroring Office/DST/the maze's terminate-on-arrival structure) cuts the deterministic violation rate $\sim 10\times$ ($\sim$20%→$\sim$2% of 150-trial cells, $k = 2, 3, 4$) but does not eliminate it: the zero counterexamples reported for Office ($k \leq 6$, Fig. 6) and the maze ($k \leq 2$), both at 150 test weights, are a rare event in a modest test set, not a guarantee Theorem 3 already rules out.*

Remark 3 (Cost)**.** *π^{hyb} needs only V_{meta} (already computed by Certify-or-Grow, Algorithm 1) plus one Bellman backup: $O(|A|d)$ per state, no k-dependence, versus $O(k|A|d)$ for $\widehat{Q}_{\mathcal{B}}$ from scratch. On real domains this asymptotic gap is real but is dominated, at the sizes tested, by V_{meta}'s own $O(|S|^3)$-per-sweep solve (*3.6–12.3*ms vs.* 136–279*ms, k=1..6, Office) — so π^{hyb} is cheap only* on top of *a system already paying for V_{meta}, not as a standalone replacement for GPI. The same reuse warm-starts Algorithm 1's fallback (line 6) from π_{meta}, needing few sweeps whenever V_{meta} is already close to V_w^*.*

Remark 4 (Validation)**.** *Zero violations of Theorem 1 in every experiment (random MDPs,* 3000 *query cases; real DST, certified k=2; real Office, canonical k=6 — `SFOLS/11_envelope_gpi.ipynb`, `SFOLS/12_envelope_gpi_dst.ipynb`, `OKB/notebooks/office_envelope_gpi.ipynb`), and Lemma 1's bound never binds (*500 *trials). Against GPI's* realized *value (a strictly stronger, unproven target): random MDPs give GPI the win in* 16/500 *and* 15/500 *trials (worst gaps* 1.42, 0.59*); real DST never loses (*189/200 *ties,* 11/200 *strict wins); real Office ties the true optimum exactly on all* 100 *weights.*

Reusing a metapolicy: the point of computing its SF, not just V_{meta}. π^{hyb} above still needs V_{meta} recomputed at *every* query weight — cheap, but it does not let a *solved* metapolicy be folded back into $\mathcal{B}$ for future queries, the way Algorithm 4 needs (§2.3): pooling π_{meta} into GPI for a future weight needs its full table $\Psi^{\pi_{\text{meta}}}(s,a) \in \mathbb{R}^d$ for *every* (s,a), not just the scalar V_{meta} — infeasible by matrix inversion once $|S|$ is large or the transition model is unknown.

Remark 5 (Correctness and coverage)**.** *The bootstrap's* arg max *ranges only over the k base policies' own actions at s' — exactly $B(s')$ — so it learns π_{meta} (Definition 2) by construction; the bootstrapped* value *plugged back in, $\widehat{\Psi}(s', \pi_{\hat{j}}(s'))$, is a full d-vector, so the same update also learns $\Psi^{\pi_{\text{meta}}}(s,a)$ at every (s,a) the behavior policy visits, whether or not $a \in B(s)$ — the same off-policy SF mechanism used to learn each base policy, requiring no transition model or linear solve. Verified on random MDPs (*10 *trials,* $n{\le}8, m{\le}5, d{\le}4, k{\le}4$, $5{\times}10^5$ *steps,* $\alpha_t = 1/(1{+}0.02 \cdot \text{visits})$*, exploring starts every* 20 *steps): the learned π_{meta} matches exact restricted PI exactly,* 0 *mismatches over every state in every trial, and $\widehat{\Psi}$ matches the exact linear solve with max error* ≤ 0.051 *over every (s,a). Without exploring starts, coverage gaps reproduce the same failure mode as* `SFOLS/09_qlearning_metapolicy.ipynb`*'s restricted learner — some (s,a) never visited, not an error in the update rule.*

The exact counterpart: merging T_B itself. With a known model, Certify-or-Grow still computes V_{meta} (scalar, via T_B) and $\Psi^{\pi_{\text{meta}}}$ (vector) as two separate solves. These merge into one synchronous recursion — but not because the natural vector lift of T_B is itself a contraction; it is not, for the same reason Theorem 3's predecessor failed.

Algorithm 3 MetaPolicyVI($\{\pi_1, \ldots, \pi_k\}, w$) — metapolicy and its SF table, exact/synchronous

$\Psi(s,j) \leftarrow \mathbf{0}$ for every s, j
repeat
  $\hat{j}(s') \leftarrow \arg\max_{j'} \Psi(s',j')^\top w$ for every s'
  $\Psi(s,j) \leftarrow \phi(s, \pi_j(s)) + \gamma\, \mathbb{E}_{s'}\big[\Psi(s', \hat{j}(s'))\big]$ for every s, j ▷ synchronous sweep, Definition 5
until converged
$\pi_{\text{meta}}(s) \leftarrow \pi_{\hat{j}(s)}(s)$ for every s
$\Psi^{\pi_{\text{meta}}}(s,a) \leftarrow \phi(s,a) + \gamma \mathbb{E}_{s'}[\Psi(s', \hat{j}(s'))]$ for every (s,a) ▷ one off-policy backup, every action
return π_{meta}, $\Psi^{\pi_{\text{meta}}}$

Remark 6 (Validation)**.** *Verified on* 15 *random MDPs ($n{\le}8, m{\le}5, d{\le}4, k{\le}4$,* 2000 *sweeps):* 0 *policy mismatches against exact restricted PI and $\widehat{\Psi}$ exact to numerical precision against the linear solve, in every trial. The greedy index (Lemma 3) stabilized within* 1–3 *sweeps in every trial — well before the γ-rate convergence Ψ itself needs.*

2.2 Comparison with Nemecek & Parr's policy caches

Certify-or-Grow's certificate (Definition 3) is exact but requires restricted policy iteration per query. Nemecek & Parr (*Policy Caches with Successor Features*, ICML 2021) address the same grow-or-reuse decision via a cheap closed-form/LP bound instead: given base policies $\{\pi_i\}$ with weights $\{w_i\}$ and a query weight w in their conical hull, their upper bound (Theorem 1) is $\min\{\sum_i \alpha_i V_i^{\pi_i}(s_0) : \sum_i \alpha_i w_i = w,\ \alpha_i \ge 0\}$ (an LP over each base policy's own optimal value in its own task), their lower bound is the standard GPI estimate $\max_a \max_i \Psi^{\pi_i}(s_0,a)^\top w$, and the cache is deemed sufficient — deploying *GPI* over the pool — iff the relative gap between the two is $\le \eta$ for a chosen threshold η (their Algorithm 1).

	fraction of 150 random query weights judged sufficient (mean / max realized suboptimality)				
	Certify-or-Grow	$\eta = 0.01$	$\eta = 0.05$	$\eta = 0.1$	$\eta = 0.2$
DST, $k = 2$	4.0% (0/0)	1.3% (0/0)	4.0% (0/0)	9.3% (0/0)	16.0% (0/0)

At $\eta = 0.3$, acceptance rises to 24.7% but realized suboptimality becomes nonzero (mean 0.0025, max 0.0524): the LP bound's slack starts to bind. Both methods reproduced faithfully and compared head-to-head on DST in `SFOLS/10_nemecek_parr_baseline.ipynb`.

Remark 7 (Neither dominates)**.** *Certify-or-Grow's exact certificate never accepts a suboptimal cache, at the cost of a per-query restricted-PI solve and a strictly rarer acceptance event (the true optimum must lie entirely inside $B(s)$ along every visited state). Nemecek & Parr's bound is cheaper per query (an LP over already-cached scalars) and, because it targets GPI's generally-better realized quality rather than the metapolicy's, stays safe over a wider acceptance region at moderate η — but offers only a tunable, occasionally-loose guarantee rather than an exact one.*

2.3 Metapolicy's own corner-weight set

Remark 8 (Restricted sub-MDP)**.** *Given a basis $\mathcal{B} = \{\pi_j\}_{j=1}^k$, let $\mathcal{M}_\mathcal{B}$ be the MDP with $\mathcal{M}$'s states, transitions, and features, but action set $B(s)$ at every state. T_B is $\mathcal{M}_\mathcal{B}$'s own Bellman optimality operator, so $V_{\text{meta}}(\cdot; w) = V_w^*$ of $\mathcal{M}_\mathcal{B}$, and $\mathcal{M}_\mathcal{B}$ has its own well-defined CCS, $\text{CCS}_\mathcal{B} \subseteq \mathbb{R}^d$, with corner weights $W_\mathcal{B}$.*

Remark 9 (The k pure points are always genuine vertices)**.** $\{\Psi^{\pi_j}(s_0, \pi_j(s_0))\}_{j=1}^k \subseteq \text{CCS}_\mathcal{B}$*, i.e. every $\Psi^{\pi_j}(s_0, \pi_j(s_0))$ survives as a non-dominated vertex of $\text{CCS}_\mathcal{B}$, however large $\mathcal{B}$ grows.*

Remark 10. *Let $h_S(w) = \max_{\psi^\pi \in S} \psi^\pi \cdot w$ then we have that:*

$$0 \leq h_{CCS}(w) - h_{CCS_\mathcal{B}}(w) \leq \frac{\max_s gap_w(s; V^\mathcal{B})}{1 - \gamma}$$

The proof is on the hand written notebook.

Algorithm 4 GET $CCS_\mathcal{B}$

$F \leftarrow \{\Psi^{\pi_j}(s_0, \pi_j(s_0))\}_{j=1}^k$, $\mathcal{P} \leftarrow \emptyset$ ▷ $\mathcal{P}$: pool of solved metapolicies, GPI-ready
$Q \leftarrow$ corner weights of $\text{conv}(F)$
while $Q \neq \emptyset$ **do**
  $w_c \leftarrow \text{pop}(Q)$
  $\pi_{\text{meta}}^{w_c}, \Psi^{\pi_{\text{meta}}^{w_c}} \leftarrow$ METAPOLICYVI$(\mathcal{B}, w_c)$ ▷ Algorithm 3: full table, not just V_{meta}
  $\psi_c \leftarrow \Psi^{\pi_{\text{meta}}^{w_c}}(s_0, \pi_{\text{meta}}^{w_c}(s_0))$
  if ψ_c not dominated by, and not already in F **then**
    $F \leftarrow F \cup \{\psi_c\}$; $\mathcal{P} \leftarrow \mathcal{P} \cup \{(\pi_{\text{meta}}^{w_c}, \Psi^{\pi_{\text{meta}}^{w_c}})\}$; push new corner weights of $\text{conv}(F)$ onto Q
  end if
end while
return F ($\text{CCS}_\mathcal{B}$), corner weights of $\text{conv}(F)$, $\mathcal{P}$

Remark 11 ($\mathcal{P}$ closes the reuse gap)**.** *Every accepted corner point now comes with its full table, so $\mathcal{P}$ (not just $\mathcal{B}$) is GPI-poolable for any future query weight (Theorem 1 and standard GPI both apply to $\mathcal{B} \cup \mathcal{P}$) — without this, F alone is a value-only front: useful for the coverage bound above, but silent on what action to take at a genuinely new weight beyond re-running Certify-or-Grow from $\mathcal{B}$ alone.*

3 SFOLS on Policy bases

Now the goal is to combine both methods in order to get a set of base policies that combined generate any policy from the CCS.

Algorithm 5 SFOLS-CERTIFIED

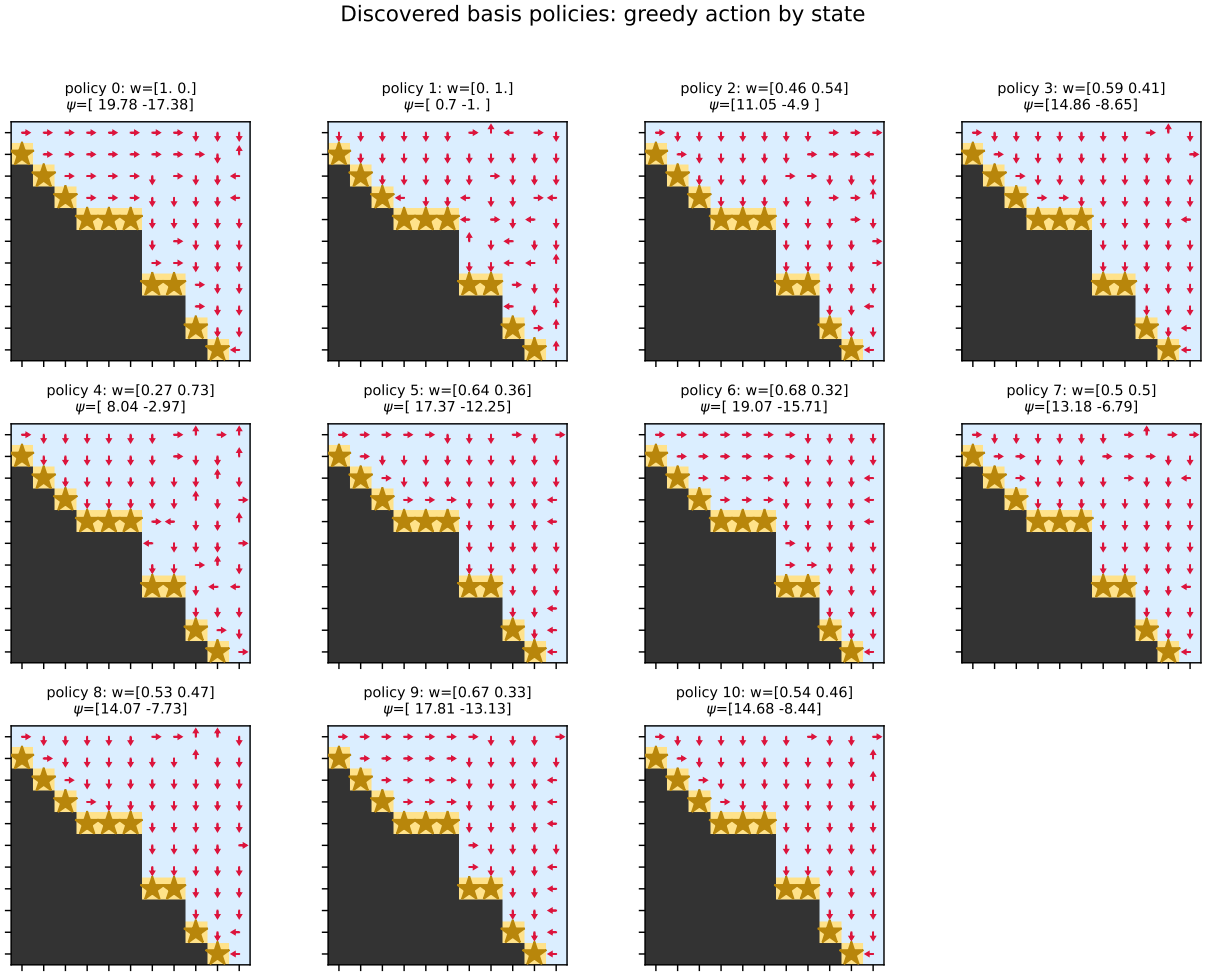
$\mathcal{B} \leftarrow \emptyset$ (policy basis), $\mathcal{F} \leftarrow \emptyset$ (value front), $\mathcal{P} \leftarrow \emptyset$ (pool, GPI-ready), $\mathcal{V} \leftarrow \emptyset$ (visited)
$Q \leftarrow \{(e_i, \infty)\}_{i=1}^{d}$
while $Q \neq \emptyset$ **and** top_priority$(Q) > \varepsilon$ **do**
    $w \leftarrow \text{pop}(Q)$, $\mathcal{V} \leftarrow \mathcal{V} \cup \{w\}$
    $\pi_{\text{meta}}, \Psi^{\pi_{\text{meta}}}, \mathcal{B} \leftarrow$ CERTIFY-OR-GROW$(w, \mathcal{B})$
    $\psi_w \leftarrow \Psi^{\pi_{\text{meta}}}(s_0, \pi_{\text{meta}}(s_0))$
    if ψ_w not dominated by, and not already in $\mathcal{F}$ **then**
        $\mathcal{F} \leftarrow \mathcal{F} \cup \{\psi_w\}$; $\mathcal{P} \leftarrow \mathcal{P} \cup \{(\pi_{\text{meta}}, \Psi^{\pi_{\text{meta}}})\}$
        $Q \leftarrow W(\text{conv}\,\mathcal{F}) \setminus \mathcal{V}$ with SFOLS priorities
    end if
end while
return $\mathcal{B}$, $\mathcal{F}$, $\mathcal{P}$

Remark 12. *Same fix as* Get-$CCS_{\mathcal{B}}$ *(Remark 11): each accepted corner now keeps its full table in $\mathcal{P}$, not just its value at s_0, so $\mathcal{P}$ (not just $\mathcal{B}$) is GPI-poolable for any future query weight.*

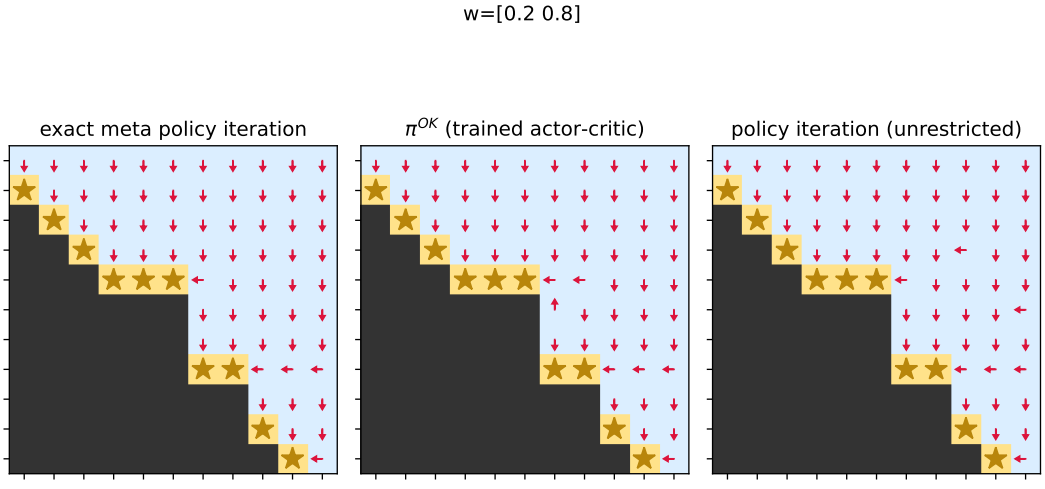
(DST $d = 2$). Both sides exact (no TD/Q-learning), 5 timed repeats, median reported.

	exact solves	median time	CCS recovered
standard SFOLS	25	382.9 ms	10/10
certified/adaptive	2 (+25 composed)	215.0 ms	10/10

Speedup **1.78**×. I think the improvement is due to the fact that we are working in at $\mathcal{M}_{\mathcal{B}}$ where the set of policies is smaller.



For a random weight $\mathbf{w} = [0.2, 0.8]$ we reach optimality:



Warm start: Most of the times an optimal policy differs very slightly from the one obtained by GPI:

$$\pi^{GPI}(s) = \arg\max_{\pi \in \Pi} \psi^{\pi}(s, a) \cdot \mathbf{w}$$

This motivates to start the Q-learning algorithm from π^{GPI} via:

$$q_w(s,a) = \max_j \psi_j^{\pi} \cdot \mathbf{w}$$

	median steps	median wall time
standard	12,500	1.241 s
warm start (GPI init)	1,500	0.144 s

Speedup **8.33**× (steps) and **8.61**× (wall time) taking median of 5 seeds each.

The metapolicy itself, learned online (`SFOLS/09_qlearning_metapolicy.ipynb`). Every result above solves the restricted operator T_B exactly (dense linear algebra). Real SFOLS/GPI implementations (`RM/sf-fsa-vi/sfols/rl/successor_features/tabular_sf.py`) learn Ψ online via TD instead. We re-implement Definition 2's restricted composition with the identical SFQL-style TD update, restricting both the ε-greedy behavior policy and the bootstrap target to $B(s)$, and compare it on DST against an unrestricted learner trained from scratch for the same query weight, same schedule, same seeds:

	250 steps	1000 steps	steps to converge
restricted metapolicy	gap = 0.000	gap = 0.000	**250**
unrestricted, from scratch	gap = 1.706	gap = 0.000	1000

4× fewer environment steps per query (mean of 10 seeds; gap to the exact optimum, DST, $w = (0.4, 0.6)$), after the $k = 2$ base pool is built once (16,000 steps, amortized exactly as in every other speedup reported here). Confirms Certify-or-Grow's restriction helps in the setting practitioners actually use, not only under exact solves.

4 FSA (non-Markovian) on Policy basis

We can grow the CCS and then perform the FSA-RM-VI algorithm. Or grow only the CCS to the size necessary to solve the FSA-RM-VI algorithm. The latter is the one we are going to present here.

Let $v = \tau(u, \mathcal{O}(\varepsilon_j))$ and write $\widehat{W}[v] = W[v]/\|W[v]\|_1$ for the normalized weight, assuming $W[v] \neq \mathbf{0}$. Then following Infante

$$W^{k+1}[u]_j = \max_{a,\, \pi \in \Pi_{\mathrm{CCS}}} \psi^{\pi}(\varepsilon_j, a)^{\top} W^k[v] \tag{1}$$

$$= \left\|W^k[v]\right\|_1 \cdot \max_{a,\, \pi \in \Pi_{\mathrm{CCS}}} \psi^{\pi}(\varepsilon_j, a)^{\top} \widehat{W}^k[v] \tag{2}$$

$$= \left\|W^k[v]\right\|_1 \cdot V^*_{\widehat{W}^k[v]}(\varepsilon_j) \tag{3}$$

$$= \left\|W^k[v]\right\|_1 \cdot \Psi^{\pi_{\mathrm{meta}}}(\varepsilon_j, \pi_{\mathrm{meta}}(\varepsilon_j))^{\top} \widehat{W}^k[v], \tag{4}$$

$$(\pi_{\mathrm{meta}}, \Psi^{\pi_{\mathrm{meta}}}, \mathcal{B}) \leftarrow \textsc{Certify-or-Grow}\big(\widehat{W}^k[v],\, \mathcal{B}\big).$$

Algorithm 6 CERTIFIED-RM-VI

$\mathcal{B} \leftarrow \emptyset$; $\mathcal{P} \leftarrow \emptyset$ ▷ pool of solved metapolicies, GPI-ready
$W[u] \leftarrow \mathbf{0} \in \mathbb{R}^{|E|}$ for all $u \in U$
repeat
  for each u, each distinct successor v of u **do**
    $E_{uv} \leftarrow \{\, j : \tau(u, \mathcal{O}(\varepsilon_j)) = v \,\}$
    if v terminal **then**
      $W[u]_j \leftarrow r(u, v)$ for $j \in E_{uv}$
    else
      $\pi_{\text{meta}}, \Psi^{\pi_{\text{meta}}}, \mathcal{B} \leftarrow \textsc{Certify-or-Grow}\big(W[v]/\|W[v]\|_1,\ \mathcal{B}\big)$
      $\mathcal{P} \leftarrow \mathcal{P} \cup \{(\pi_{\text{meta}}, \Psi^{\pi_{\text{meta}}})\}$ ▷ same fix as Remark 11: cache the full table, not just the exit value
      $W[u]_j \leftarrow \|W[v]\|_1 \cdot \Psi^{\pi_{\text{meta}}}(\varepsilon_j, \pi_{\text{meta}}(\varepsilon_j))^\top (W[v]/\|W[v]\|_1)$ for $j \in E_{uv}$
    end if
  end for
until $\|W - W_{\text{prev}}\|_\infty \le \varepsilon$
for each u **do**
  $\pi_{\text{meta}}, \Psi^{\pi_{\text{meta}}}, \mathcal{B} \leftarrow \textsc{Certify-or-Grow}\big(W[u]/\|W[u]\|_1,\ \mathcal{B}\big)$
  $\mathcal{P} \leftarrow \mathcal{P} \cup \{(\pi_{\text{meta}}, \Psi^{\pi_{\text{meta}}})\}$
end for
return W, $\mathcal{B}$, $\mathcal{P}$

Remark 13 (Coverage gap)**.** *The sweep only ever queries $W[v]$ at edge* targets*. The start state is never certified during the loop, even though the acting policy at u_0 still needs $V^{\pi(W[u_0])}$ to be exact, we fix it in the post certification.*

Results (Office domain, "coffee → mail → office")

	basis size	sweeps	median time
standard (SFOLS discovers CCS, then traverse)	17 exact solves	4	40886.4 ms
certified/adaptive, exact (ours)	3 exact solves (3 distinct)	4	2997.9 ms

The speed up in this algorithm is of about **13.64**×.

Remark 14 (Domain-specific basis size)**.** *Office's domain only needs the $d = 6$ canonical corners to perform Infante's algorithm succesfully. Our adaptative basis reduces this to* 3 *distinct corners, however it will need to compute other policies if the higher MDP is rewriten.*

To do:

- Backpropagation of Certified-RM-VI can be optimized.
- What happens with continous MORL?
- What happens with other high-level tasks.

4.1 Exact one-step adjacency characterizes the $k = 1$ gap

Proposition 1 (Single-policy case)**.** *Single-pickup-terminal MO-MDP with reward states $R_1, \ldots, R_d$ ($\phi(s, a) = e_i$ iff a leads directly to R_i, terminal on arrival, 0 otherwise). Let $\mathcal{B} = \{\pi_1\}$, π_1 exactly optimal for e_1. If no state on π_1's trajectory from s is one step from R_j for any $j \neq 1$, then $\pi^{\text{GPI}}_{\mathcal{B}}(s; w) = \pi_{\text{meta}}(s; w) = \pi_1(s)$ for every w with $w_1 > 0$.*

Proof. $B(s) = \{\pi_1(s)\}$, so $\pi_{\text{meta}}(s) = \pi_1(s)$ trivially. For GPI: π_1's own continuation from any state never visits another reward (same hypothesis, applied recursively), so $\Psi^{\pi_1}(s', \pi_1(s')) = c(s')e_1$ for some $c(s') \geq 0$. No action from s reaches R_j, $j \neq 1$, in one step, so $\phi(s, a)^\top w \in \{0, w_1\}$ for every a, giving $\Psi^{\pi_1}(s, a)^\top w = w_1 \cdot \left[\mathbb{1}(a \text{ reaches } R_1) + \gamma c(s'(a))\mathbb{1}(\text{else})\right] = w_1 \cdot \Psi^{\pi_1}(s, a)^\top e_1$, a positive ($w_1 > 0$) rescaling of $\Psi^{\pi_1}(s, a)^\top e_1$, hence maximized at the same $a = \pi_1(s)$ that maximizes the latter, by definition of π_1's own optimality. □

Confirmed by direct sweep (`SFOLS/07_proximity_sweep.ipynb`): $\pi^{\text{GPI}}_{\mathcal{B}} = \pi_{\text{meta}}$ exactly for every tested adjacency distance $d \geq 2$, with a strict, nonzero gap appearing only at $d = 1$.

Measured cost of iterating π^{hyb} to full optimality (Office, canonical-corner basis $k = 1, \ldots, 6$, 30 test weights). Each k warm-starts the same full-action-set policy-iteration loop at π_{meta}'s own greedy policy and counts sweeps to the fixed point, against a cold start (`policy`$= 0$ everywhere, Line 6's current behavior):

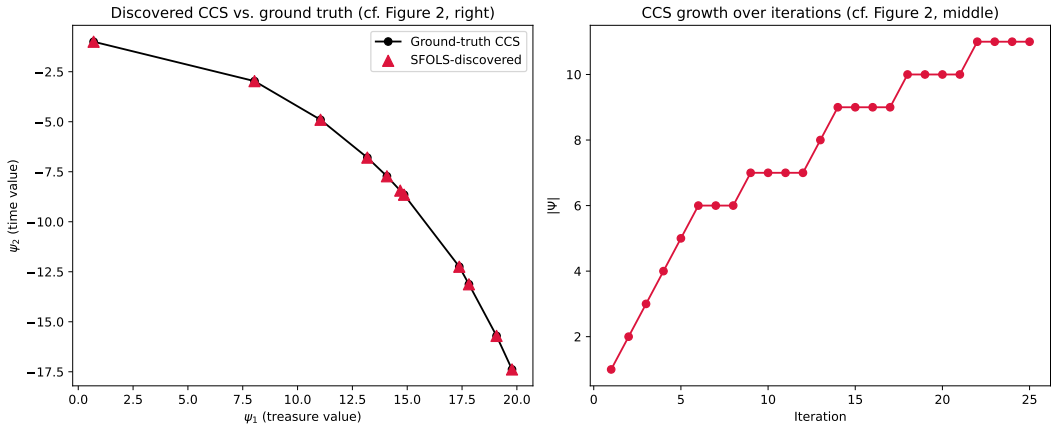
	mean sweeps	min / max sweeps	speedup vs. cold
cold start (current fallback)	16.13	10 / 21	—
$k = 1$	14.10	2 / 24	1.14×
$k = 2$	12.87	2 / 24	1.25×
$k = 3$	12.07	2 / 24	1.34×
$k = 4$	9.93	2 / 24	1.62×
$k = 5$	7.03	2 / 24	2.29×
$k = 6$	3.83	2 / 10	**4.21**×

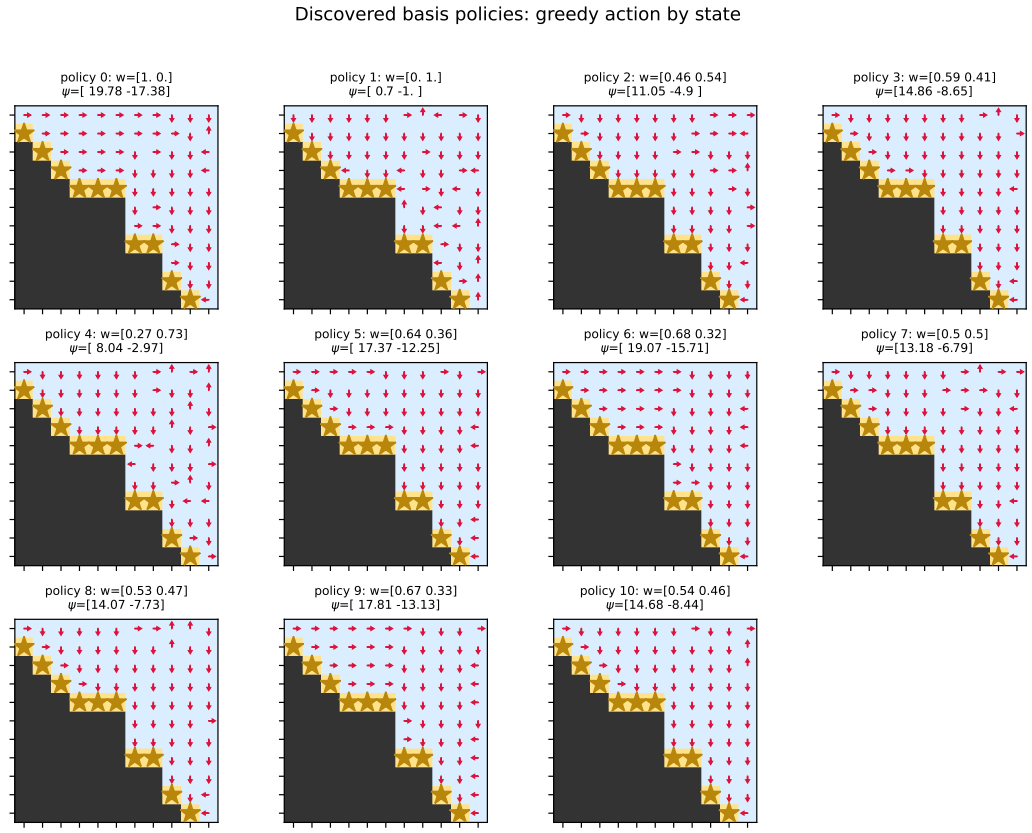
Mean cost drops monotonically as the basis grows: a 4.21× reduction in exact-solve sweeps at the complete basis, consistent in kind with the 8.33× Q-learning warm-start speedup reported above (p.10). Two caveats keep this honest rather than optimistic:

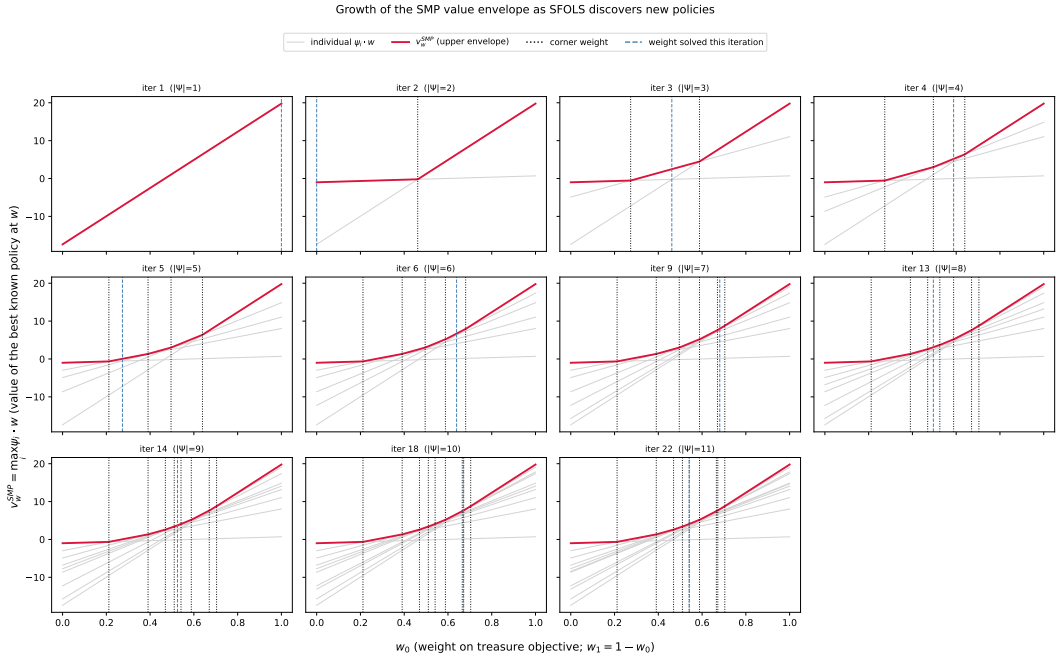
Remark 15 (The "already stable" certificate never fired, even at $k = 6$)**.** *Not one of the 30 weights reached $\pi^{\text{hyb}} = \pi_{\text{meta}}$'s own action in a single sweep, even though $k = 6$'s basis makes $V_{\text{meta}} = V^*_w$ exactly in this domain. Office's terminate-on-any-object structure produces many tied Q-values, and* arg max *breaks ties by lowest index — so the freshly-recomputed full-action greedy action can differ in* label *from π_{meta}'s own choice while being equal in* value*, costing one wasted-but-value-neutral extra sweep. The real certificate is the Bellman-gap test (as in `meta_policy_iteration_certified`), not exact policy equality — the gap is zero at $k = 6$ even though the sweep count never bottoms out at exactly* 1*.*

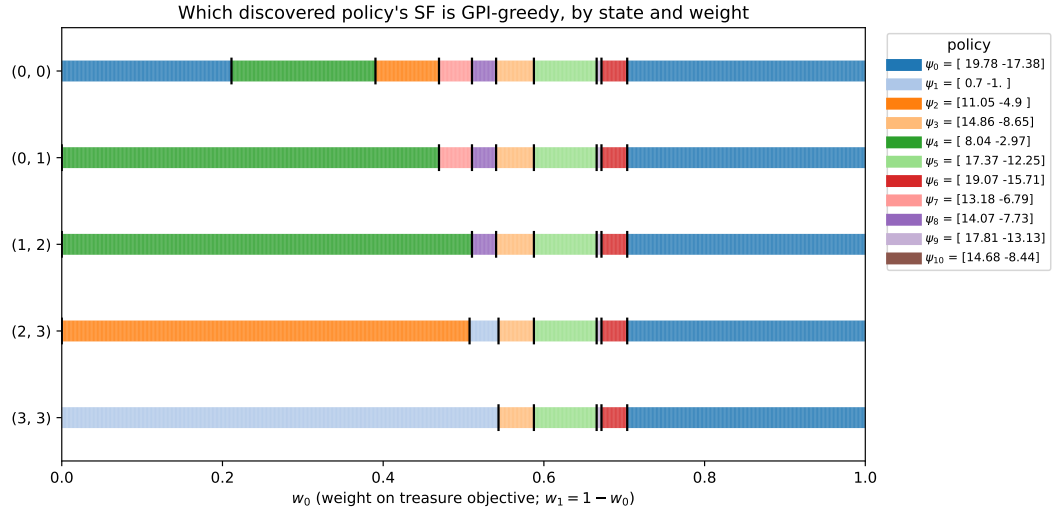
Remark 16 (Worst-case sweeps are not monotone in k)**.** *The maximum over the 30 weights stays at* 24 *— worse than the cold start's own worst case of* 21 *— for every $k \leq 5$, dropping to* 10 *only at $k = 6$. A value estimate numerically close to V^*_w does not guarantee its greedy policy is close in* policy space*: tie-heavy domains like Office can thrash between equally-good actions for several rounds before settling. So the warm start is a better bet* on average*, not a per-instance guarantee.*

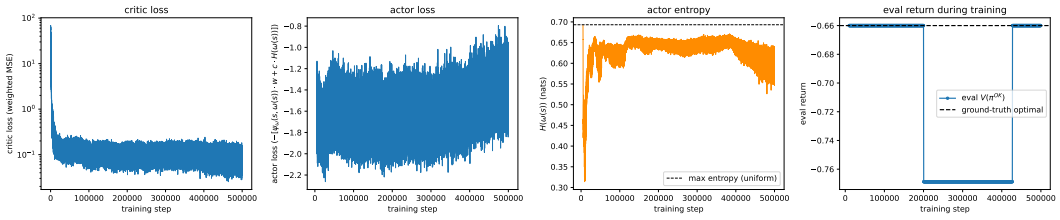
5 DST


Discovered CCS vs. ground truth (cf. Figure 2, right)
Ground-truth CCS
SFOLS-discovered
ψ_2 (time value)
ψ_1 (treasure value)
CCS growth over iterations (cf. Figure 2, middle)
|Ψ|
Iteration

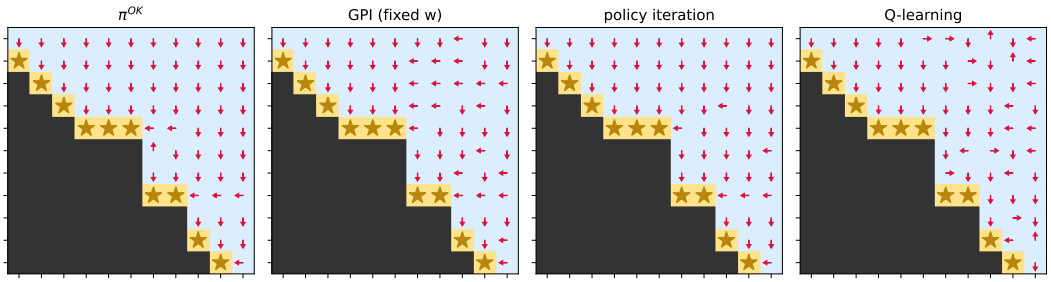

Discovered basis policies: greedy action by state
policy 0: w=[1. 0.]
ψ=[19.78 -17.38]
policy 1: w=[0. 1.]
ψ=[0.7 -1.]
policy 2: w=[0.46 0.54]
ψ=[11.05 -4.9]
policy 3: w=[0.59 0.41]
ψ=[14.86 -8.65]
policy 4: w=[0.27 0.73]
ψ=[8.04 -2.97]
policy 5: w=[0.64 0.36]
ψ=[17.37 -12.25]
policy 6: w=[0.68 0.32]
ψ=[19.07 -15.71]
policy 7: w=[0.5 0.5]
ψ=[13.18 -6.79]
policy 8: w=[0.53 0.47]
ψ=[14.07 -7.73]
policy 9: w=[0.67 0.33]
ψ=[17.81 -13.13]
policy 10: w=[0.54 0.46]
ψ=[14.68 -8.44]


Growth of the SMP value envelope as SFOLS discovers new policies
individual $\psi_i \cdot w$
V_w^{SMP} (upper envelope)
corner weight
weight solved this iteration
iter 1 (|Ψ|=1)
iter 2 (|Ψ|=2)
iter 3 (|Ψ|=3)
iter 4 (|Ψ|=4)
iter 5 (|Ψ|=5)
iter 6 (|Ψ|=6)
iter 9 (|Ψ|=7)
iter 13 (|Ψ|=8)
iter 14 (|Ψ|=9)
iter 18 (|Ψ|=10)
iter 22 (|Ψ|=11)
$V_w^{SMP} = \max_i \psi_i \cdot w$ (value of the best known policy at w)
w_0 (weight on treasure objective; $w_1 = 1 - w_0$)

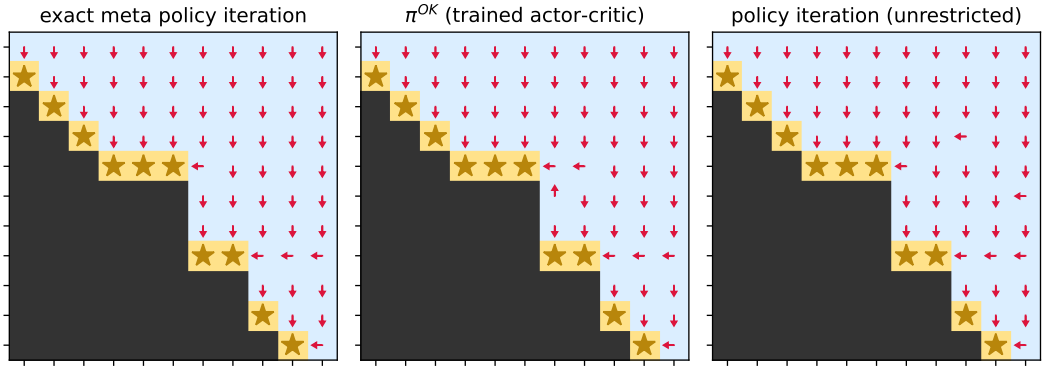

Which discovered policy's SF is GPI-greedy, by state and weight
(0, 0)
(0, 1)
(1, 2)
(2, 3)
(3, 3)
policy
ψ_0 = [19.78 -17.38]
ψ_1 = [0.7 -1.]
ψ_2 = [11.05 -4.9]
ψ_3 = [14.86 -8.65]
ψ_4 = [8.04 -2.97]
ψ_5 = [17.37 -12.25]
ψ_6 = [19.07 -15.71]
ψ_7 = [13.18 -6.79]
ψ_8 = [14.07 -7.73]
ψ_9 = [17.81 -13.13]
ψ_{10} = [14.68 -8.44]
w_0 (weight on treasure objective; $w_1 = 1 - w_0$)

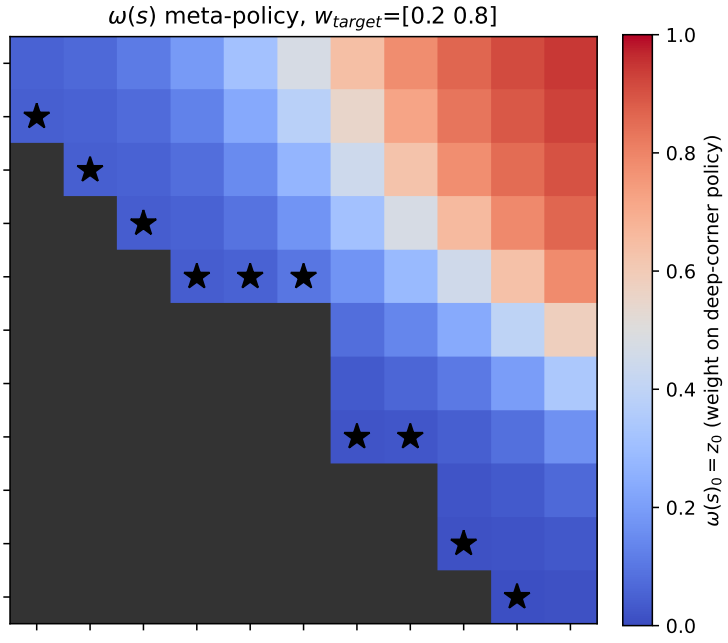

critic loss
critic loss (weighted MSE)
training step
actor loss
actor loss ($-\hat{q}_{\psi,\theta}(s,\pi(s)) \cdot w + c \cdot H_{\theta}(\pi(s))$)
training step
actor entropy
$H_{\theta}(\pi(s))$ (nats)
max entropy (uniform)
training step
eval return during training
eval return
eval $V(\pi^{GK})$
ground-truth optimal
training step

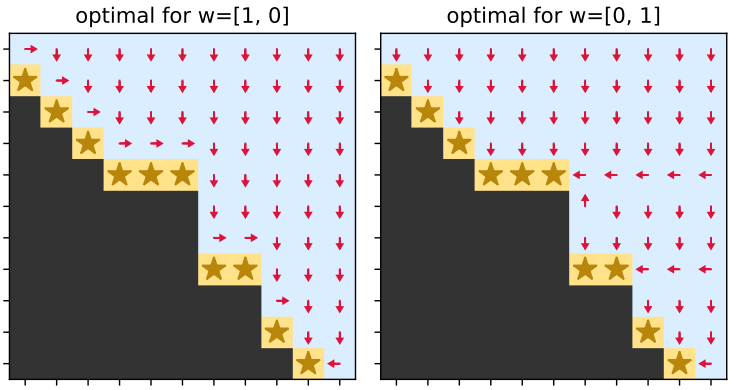
w=[0.2 0.8]

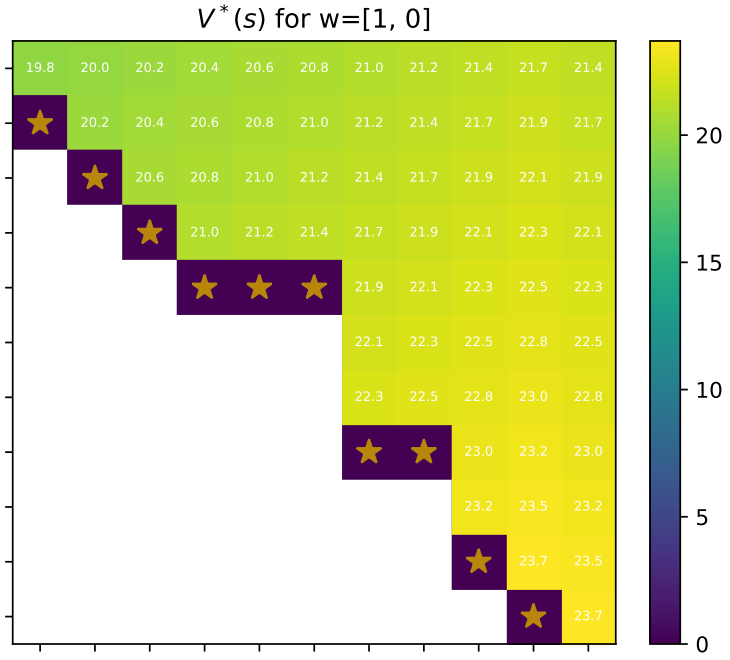

π^{OK}
GPI (fixed w)
policy iteration
Q-learning

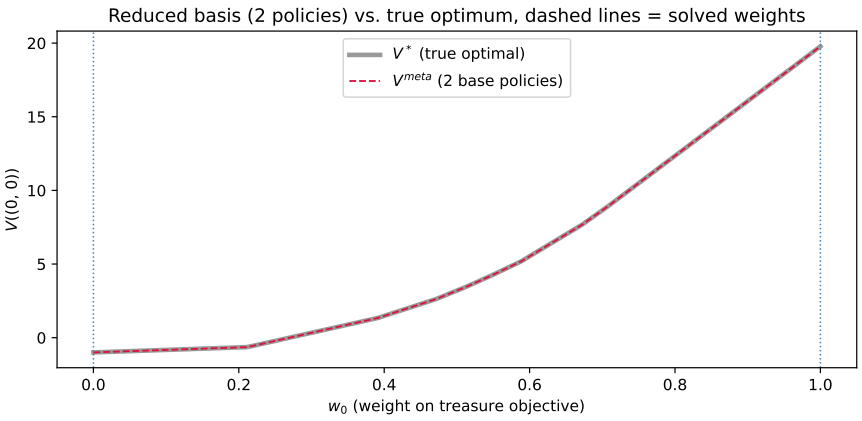
w=[0.2 0.8]


exact meta policy iteration
π^{OK} (trained actor-critic)
policy iteration (unrestricted)

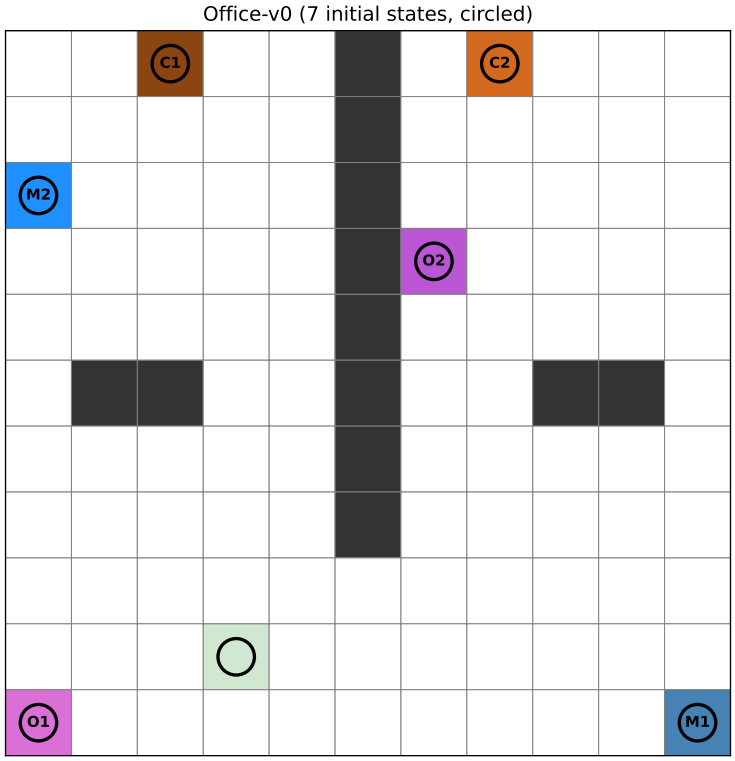

$\omega(s)$ meta-policy, w_{target}=[0.2 0.8]
1.0
0.8
0.6
0.4
0.2
0.0
$\omega(s)_0 = z_0$ (weight on deep-corner policy)

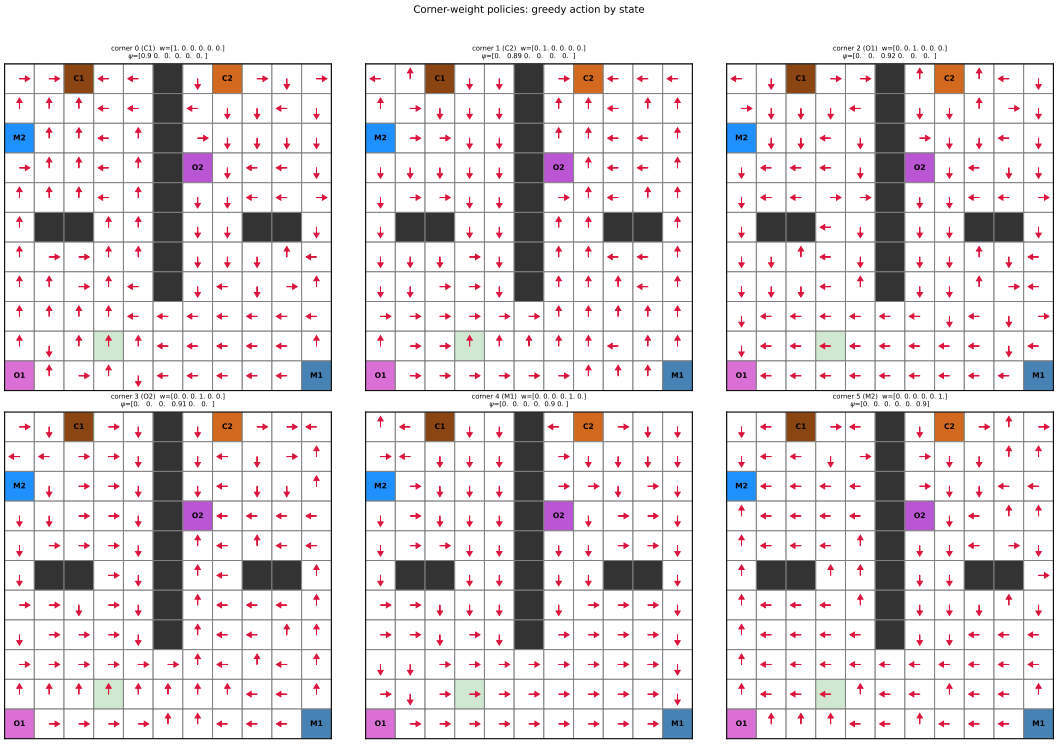

optimal for w=[1, 0]
optimal for w=[0, 1]

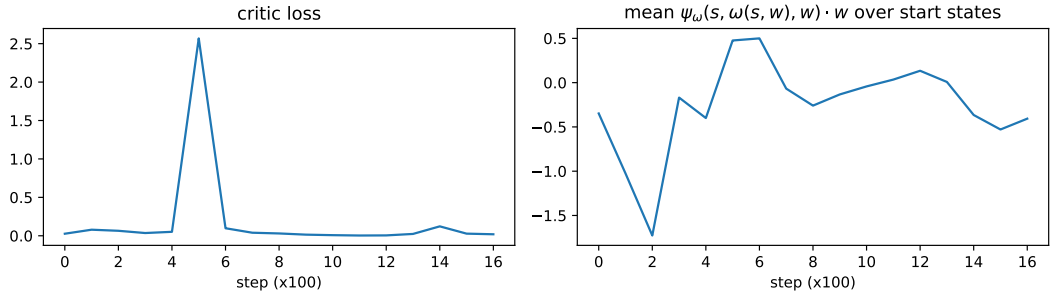

$V^*(s)$ for w=[1, 0]
19.8 20.0 20.2 20.4 20.6 20.8 21.0 21.2 21.4 21.7 21.4
20.2 20.4 20.6 20.8 21.0 21.2 21.4 21.7 21.9 21.7
20.6 20.8 21.0 21.2 21.4 21.7 21.9 22.1 21.9
21.0 21.2 21.4 21.7 21.9 22.1 22.3 22.1
21.9 22.1 22.3 22.5 22.3
22.1 22.3 22.5 22.8 22.5
22.3 22.5 22.8 23.0 22.8
23.0 23.2 23.0
23.2 23.5 23.2
23.7 23.5
23.7
20
15
10
5
0

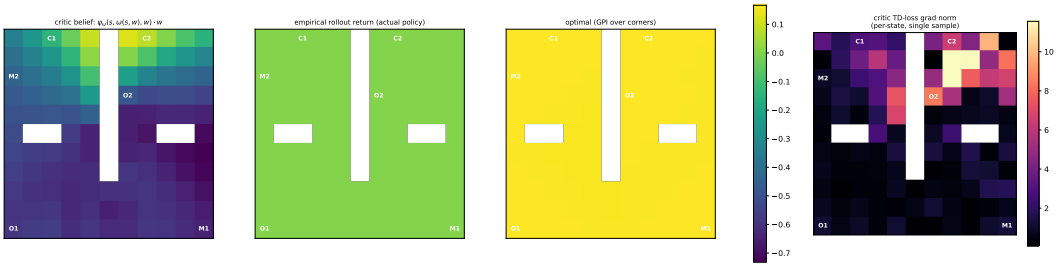

Reduced basis (2 policies) vs. true optimum, dashed lines = solved weights
V^* (true optimal)
V^{meta} (2 base policies)
$V((0, 0))$
20
15
10
5
0
0.0
0.2
0.4
0.6
0.8
1.0
w_0 (weight on treasure objective)

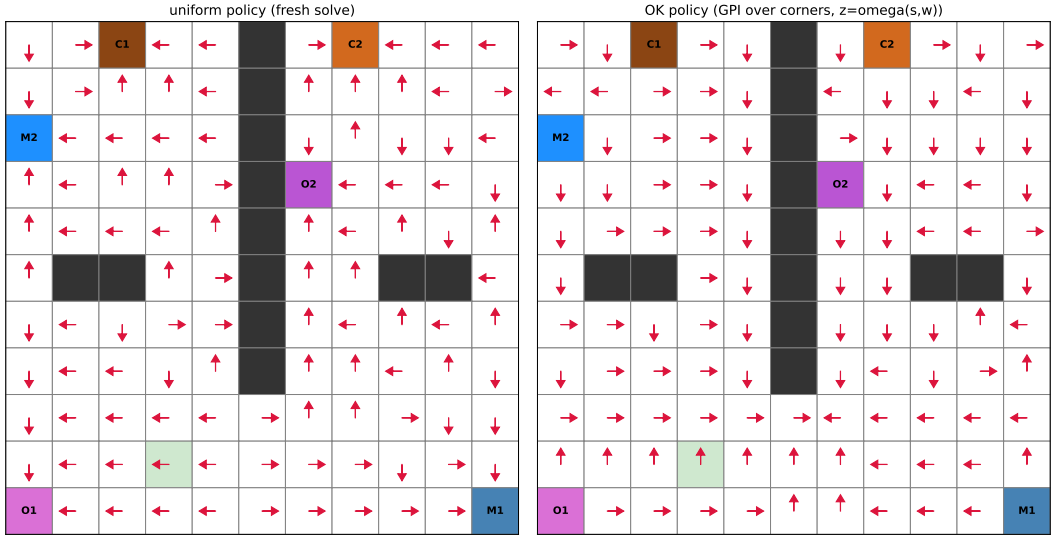
6 Office

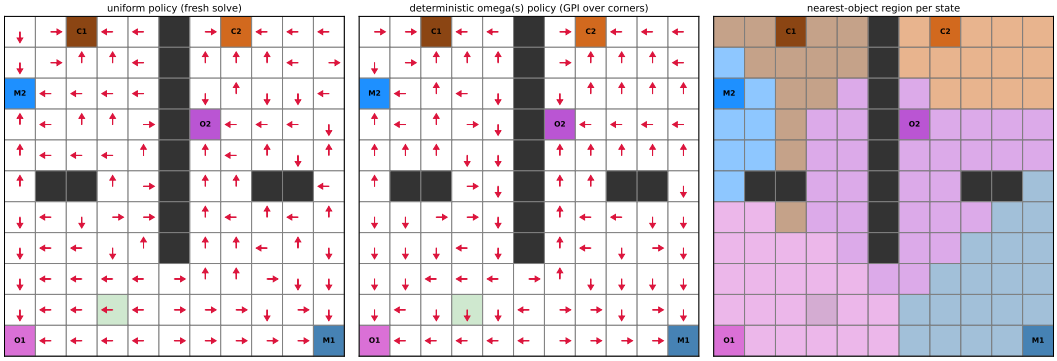

Office-v0 (7 initial states, circled)
C1
C2
M2
O2
O1
M1

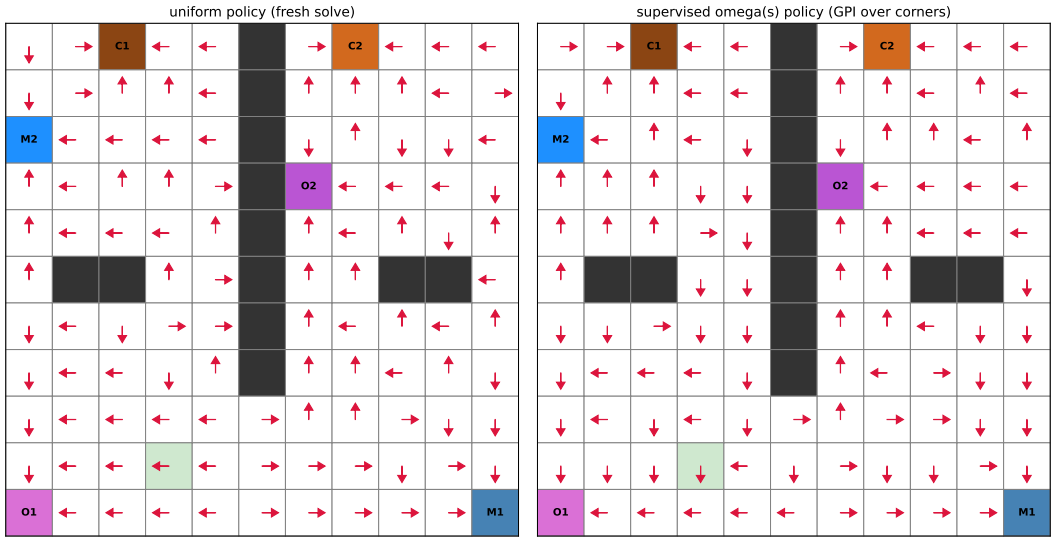

Corner-weight policies: greedy action by state
corner 0 (C1) w=[1. 0. 0. 0. 0. 0.]
corner 1 (C2) w=[0. 1. 0. 0. 0. 0.]
corner 2 (O1) w=[0. 0. 1. 0. 0. 0.]
corner 3 (O2) w=[0. 0. 0. 1. 0. 0.]
corner 4 (M1) w=[0. 0. 0. 0. 1. 0.]
corner 5 (M2) w=[0. 0. 0. 0. 0. 1.]

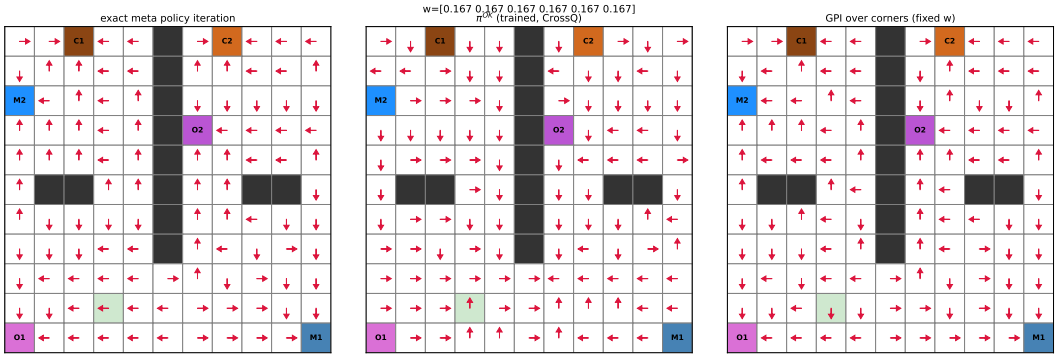

critic loss
step (x100)
mean $\psi_\omega(s, \omega(s,w), w) \cdot w$ over start states
step (x100)

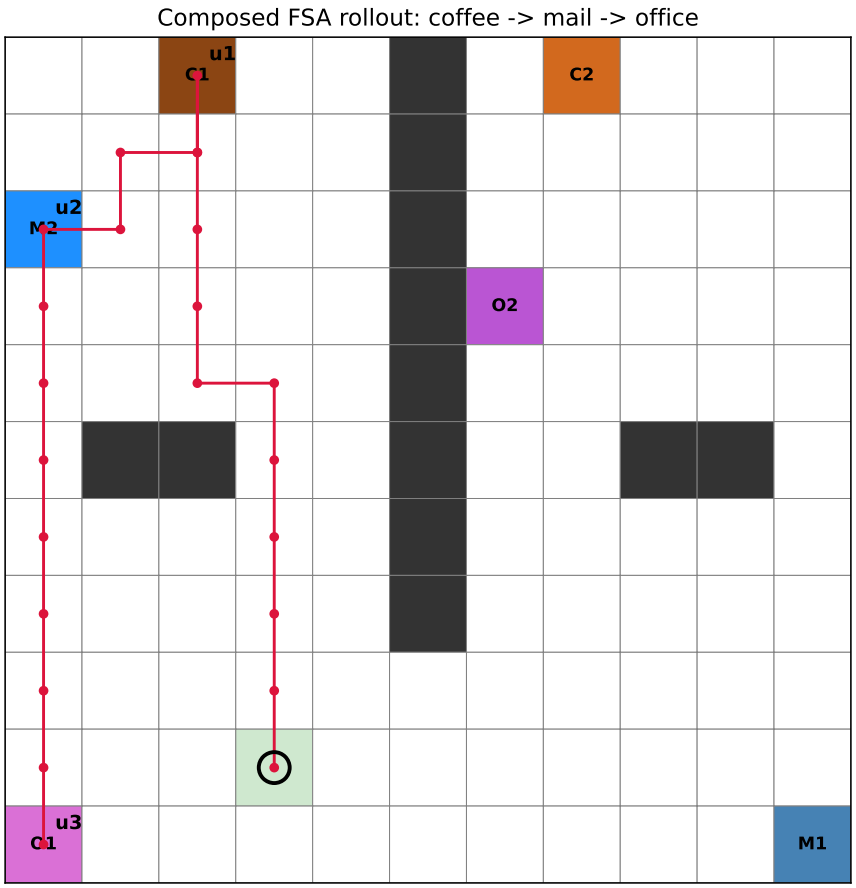

critic belief: $\psi_\omega(s, \omega(s,w), w) \cdot w$
empirical rollout return (actual policy)
optimal (GPI over corners)
critic TD-loss grad norm
(per-state, single sample)
C1
C2
M2
O2
O1
M1


uniform policy (fresh solve)
OK policy (GPI over corners, z=omega(s,w))
C1
C2
M2
O2
O1
M1


uniform policy (fresh solve)
deterministic omega(s) policy (GPI over corners)
nearest-object region per state
C1
C2
M2
O2
O1
M1


uniform policy (fresh solve)
supervised omega(s) policy (GPI over corners)
C1
C2
M2
O2
O1
M1


exact meta policy iteration
w=[0.167 0.167 0.167 0.167 0.167 0.167]
π^{ck} (trained, CrossQ)
GPI over corners (fixed w)
C1
C2
M2
O2
O1
M1


Composed FSA rollout: coffee -> mail -> office
u1
C1
C2
u2
M2
O2
u3
C1
M1